PHYSICS 101 NO: _____

NAME: _____

PHYS101/Summer 2020 — MidTerm Exam

(On my honor I have neither given nor received any unauthorized assistance on this exam.)

- This is a closed-everything exam.
- Calculators or computing devices are neither allowed nor needed.
- Exam duration is **80 minutes**.
- Consists of <u>three</u> problems worth of **100 points**.
- Answers without work shown or proper explanation will not be given any credit.

(Leave blank)

Problem 1	Problem 2	Problem 3	Problem 4	Total

Formula Sheet

Unit Vector Notation:

$\vec{A} = A_x\hat{\imath} + A_y\hat{\jmath} + A_z\hat{k}$

Scalar Product:

$\vec{A}\cdot\vec{B} = A_xB_x + A_yB_y + A_zB_z$

$\vec{A}\cdot\vec{B} = |A||B|\cos\phi$

Vector Product:

$\vec{A}\times\vec{B} = \vec{C}$

$C_x = A_yB_z - A_zB_y$

$C_y = A_zB_x - A_xB_z$

$C_z = A_xB_y - A_yB_x$

$|\vec{A}\times\vec{B}| = |A||B|\sin\phi$

Motion in three dimensions:

$\vec{r}(t) = x(t)\hat{\imath} + y(t)\hat{\jmath} + z(t)\hat{k}$

$\vec{v}_{av} = \dfrac{\vec{r}_2 - \vec{r}_1}{t_2 - t_1} = \dfrac{\boldsymbol{\Delta\vec{r}}}{\boldsymbol{\Delta t}}$

$\vec{v}(t) = \lim\limits_{\Delta t\to 0}\dfrac{\boldsymbol{\Delta\vec{r}}}{\boldsymbol{\Delta t}} = \dfrac{d\vec{r}}{dt}$

$\vec{a}_{av} = \dfrac{\vec{v}_2 - \vec{v}_1}{t_2 - t_1} = \dfrac{\boldsymbol{\Delta\vec{v}}}{\boldsymbol{\Delta t}}$

$\vec{a}(t) = \lim\limits_{\Delta t\to 0}\dfrac{\boldsymbol{\Delta\vec{v}}}{\boldsymbol{\Delta t}} = \dfrac{d\vec{v}}{dt}$

Constant x-acceleration:

$v_x(t) = v_{0x} + a_xt$

$x(t) = x_0 + v_{0x}t + \dfrac{1}{2}a_xt^2$

$v_x^2 = v_{0x}^2 + 2a_x(x - x_0)$

Varying acceleration:

$\vec{a} = \vec{a}(t)$, is a func. of time

$\vec{v}(t) = \int_0^t \vec{a}(t)dt + \vec{v}_0$

$\vec{r}(t) = \int_0^t \vec{v}(t)dt + \vec{r}_0$

Projectile Motion:

$a_x = 0, \quad a_y = -g$

Uniform Circular Motion:

$a_{\text{radial}} = v^2/R, \quad \vec{v}\perp\vec{a}_{\text{rad}}$

Period: $T = 2\pi R/v$

Motion Along a Curve:

$a_{\text{radial}} = v^2/R$

$a = \sqrt{a_\perp^2 + a_\parallel^2}$

Newton's 1st Law:

$\vec{R} = \vec{F}_1 + \vec{F}_2 + \ldots = \sum\vec{F}$

$\sum\vec{F} = 0$

Newton's 2nd Law:

$\sum\vec{F} = m\vec{a}$

Newton's 3rd Law:

$\vec{F}_{\text{A on B}} = -\vec{F}_{\text{B on A}}$

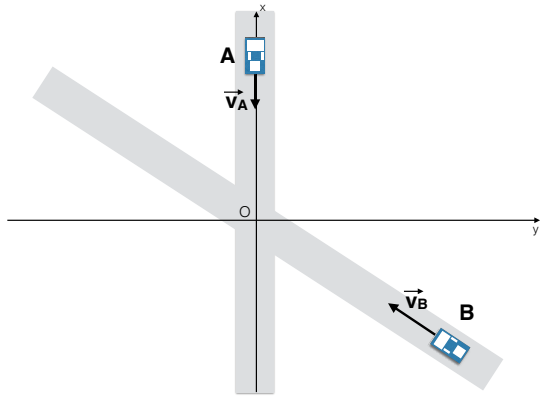
Friction Force:

$f_k = \mu_k n \quad f_s \le \mu_s n$

$\sin 36.9^\circ = \cos 53.1^\circ = 0.6$, $\sin 53.1^\circ = \cos 36.9^\circ = 0.8$, $\sin 45^\circ = \cos 45^\circ = \sqrt{2}/2$

Problem (1) (25 points)

At the instant shown, two cars are approaching an intersection of two highways. The position functions of the two cars are given as follows:



Car A:

$$\vec{r}_A(t) = \Big(200\text{ m} - (70\text{ m/s})t + (5\text{ m/s}^2)t^2\Big)\hat{\jmath}$$

Car B:

$$\vec{r}_B(t) = \Big(300\text{ m} - (30\text{ m/s})t\Big)\hat{\imath} + \Big(-400 + (40\text{ m/s})t\Big)\hat{\jmath}$$

taking the intersection point as the origin.

(a) (10 pts.) Find the distance of Car B from the intersection when Car A first reaches there.

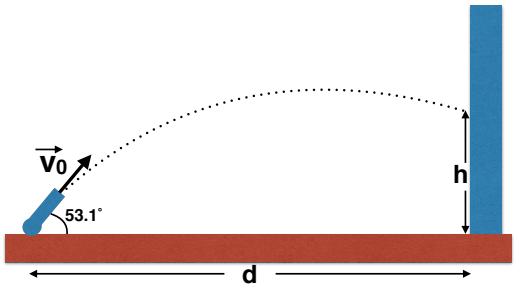
$d=$

(b) (15 pts.) Find the velocity of Car B with respect to Car A at $t = 2$ s in unit-vector notation.

$\vec{v}_{B/A}=$

Problem (2) (25 points)

A ball is thrown toward a wall at speed $v_0 = 30$ m/s and at angle 53.1° above the horizontal. The wall is at distance $d = 54$ m from the release point. (Take $g = 10$ m/s^2; $\sin\ 53.1° = 0.8$; $\cos\ 53.1° = 0.6$)



(a) (10 pts.) How far above the release point does the ball hit the wall?

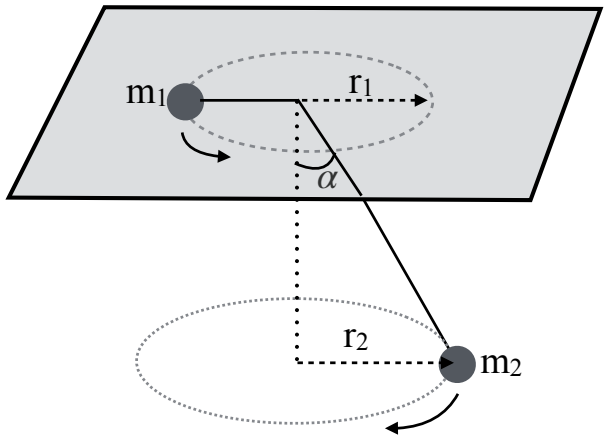
$h=$

(b) (15 pts.) Find the velocity of the ball as it hits the wall in <u>unit vector notation</u>.

$\vec{v}=$

Problem 3 (25 points)

An object with mass $m_1 = 1.5$ kg slides on a smooth, frictionless table. The object is constrained to move in a circle by a string that passes through a hole in the center of the table and is attached to a second object with the same mass $m_2 = 0.6$ kg hanging below the table. The second object swings in a circle in air, so that the string makes an angle α with the vertical. The radius of the circular motion of the objects are $r_1 = 0.040$ m and $r_2 = 0.075$ m. The period of the circular motion of m_2 is $T_2 = 0.6$ s. (Take $g \simeq 10$ m/s^2 and $\pi \simeq 3$)



(a) (5 pts.) Draw the free-body diagram of the two objects. Show the direction of acceleration for each object.

(b) (10 pts.) Find the angle α.

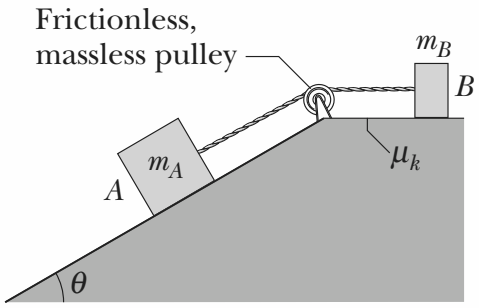
$\theta =$

(c) (10 pts.) Find the radial acceleration of the object (m_1) on the table.

$a_{\mathbf{rad}} =$

Problem (4) (25 points)

Block A in the figure has mass m_A=4.0 kg, and block B has mass m_B=2.0 kg. The coefficient of kinetic friction between block B and the horizontal plane is $\mu_k = 0.50$. The block A moves downward on a frictionless inclined plane with an inclination angle of $\theta = 30°$. The pulley serves only to change the direction of the cord connecting the blocks. The cord has negligible mass and you may take g=10m/s^2.



(a) (5 pts.) Draw the free-body diagram of each object.

(b) (10 pts.) Find the tension in the cord.

$T=$

(c) (10 pts.) Find the magnitude of the acceleration of the blocks.

$a=$